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Probing the Transport of Ionic Liquids in Aqueous Solution through Nanopores

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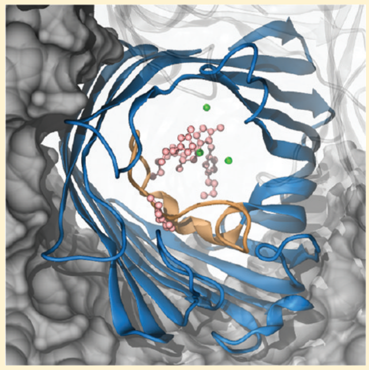
Supporting Information

ABSTRACT: The temperature-dependent transport of the ionic liquid 1-butyl-3-methylimidazolium chloride (BMIM-Cl) in aqueous solution is studied theoretically and experimentally. Using molecular dynamics simulations and ion-conductance measurements, the transport is examined in bulk as well as through a biological nanopore, that is, OmpF and its mutant D113A. This investigation is motivated by the observation that aqueous solutions of BMIM-Cl drastically reduce the translocation speed of DNA or antibiotics through nanopores in electrophysiological measurements. This makes BMIM-Cl an interesting alternative salt to improve the time resolution. In line with previous investigations of simple salts, the size of the ions and their orientation adds another important degree of freedom to the ion transport, thereby slowing the transport through nanopores. An excellent agreement between theory and conductance measurements is obtained for wild type OmpF and a reasonable agreement for the mutant. Moreover, all-atom simulations allow an atomistic analysis revealing molecular details of the rate-limiting ion interactions with the channel.



SECTION: Biophysical Chemistry

Pure ionic liquids (ILs), usually molten salts with a melting point lower than 100 °C, have become a subject of intense studies in recent years. ILs are characterized by a high ion conductivity, thermal and chemical stability, as well as favorable solvation properties. These characteristics of ILs generated enormous interest, and they can be applied to a large number of useful applications including organic synthesis, catalysis, separation, and electrochemical studies.[1−5] Moreover, mixtures of water and ILs, that is, ILs in aqueous solution, have been investigated recently,[6,7] which is actually the subject of our study.

Ion and substrate transport through nanopores is currently an active field of research with exciting biological applications.[8,9] Concerning the ion transport, most studies, especially including the effect of temperature, have been restricted to simple ions such as KCl, NaCl, or $MgCl_2$ (see, e.g., refs 10−18). Concerning substrate transport, one investigates, for example, the translocation of antibiotics,[19] DNA,[9] and peptides[20] through nanopores. The translocation of these substrates is experimentally studied using electrophysiology, that is, the use of ion-current measurements as an indirect probe. One of the potential applications of IL solutions is slowing down the kinetics of fast permeation events through nanopores. To this end, de Zoysa et al.[21] successfully used a 1 M 1-butyl-3-methyl-imidazolium chloride (BMIM-Cl) (Figure 1a) solution to slow down the DNA translocation in a nanopore. This reduction in speed might help in DNA sequencing by increasing the residence time of DNA in the pore.[21] Furthermore, we have recently shown that a 1 M BMIM-Cl solution increases the residence time of ampicillin in the outer membrane porin F (OmpF) pore by a factor of 3 compared with a KCl solution of the same concentration.[22] Again, this increase in residence time improved the time resolution to catch fast events of antibiotic translocation through the pore, which is otherwise hard to detect or even undetectable. In the same study, it was found that BMIM-Cl compared with KCl leads to a roughly two-fold reduction in bulk conductivity, whereas the respective OmpF conductance decreases by a factor of ∼7.[22] This study is devoted to understand the underlying interactions of ILs with the nanopore walls.

The investigated nanopore is the general diffusion porin OmpF, that is, a pore in the outer membrane of the *E. coli* bacterium. (See Figure 1c.) High-resolution crystal structures of OmpF reveal their trimeric organization.[23,24] The constriction zone contains three positively charged arginine residues R42, R82, and R132 on the one side and two negatively charged residues, namely, aspartic acid D113 and glutamic acid E117, on the other side. (See Figure S1in the Supporting Information.) This spatial arrangement of positively and negatively charged residues is responsible for a strong transverse electric field inside the pore leading to particular orientations of molecules in this region.[25−27] The inner loop L3 is a part of the constriction zone and has a large influence on the translocation properties of the pore.

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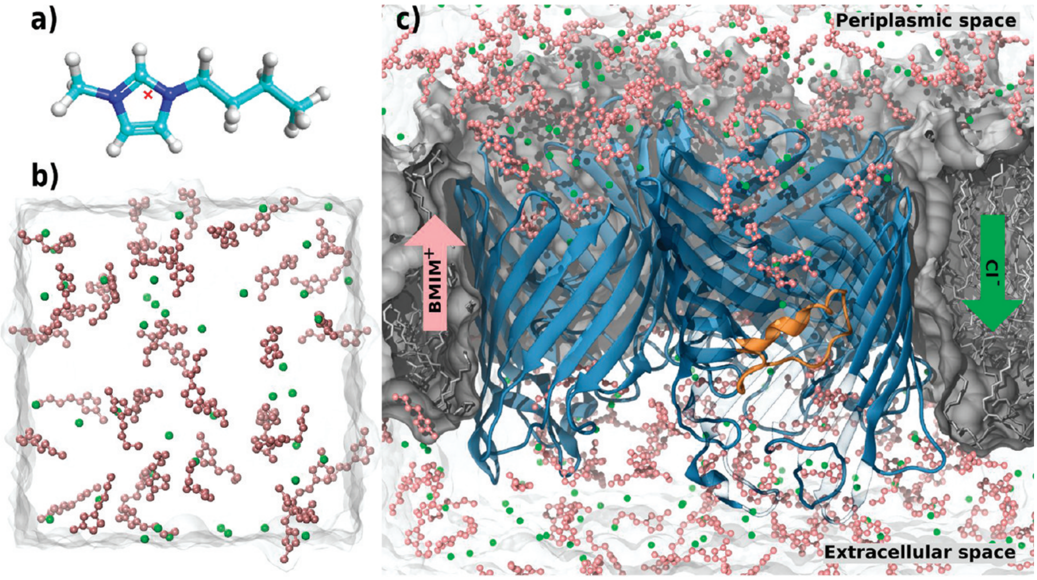

Figure 1. (a) Structure of $BMIM^+$ (atom colors: C, cyan; N, blue; H, white) including the center of charge indicated by a red cross. (b) Simulation box for the bulk simulations. The Cl^- ions are shown as green spheres and the $BMIM^+$ ions are shown as pink ball-stick structures. (c) Simulation setup for the pore simulations including OmpF (blue, secondary structure; orange, loop L3) and the membrane (gray surface and stick representation). The arrows indicate the direction of the movement of ions under the effect of a positive applied field (in the same direction as the flow of the $BMIM^+$ ions). For the sake of visualization, part of the monomer in the front has been removed. The figures were generated using VMD.[28]

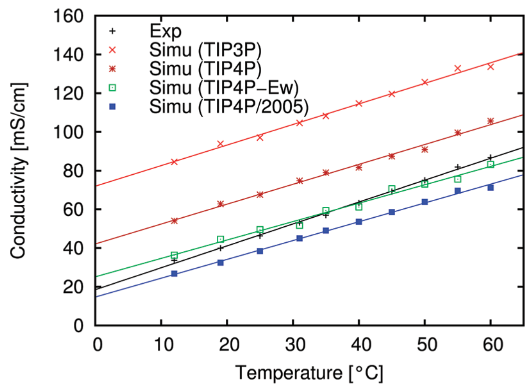

Figure 2. Bulk conductivity versus temperature for different water models compared to experiments.

In this study, we investigated molecular details of BMIM-Cl transport through OmpF. In particular, the behavior and orientation of the bulky ions within the channel and the constriction region are of interest. To this end, we performed applied field MD simulations to model the ion flow of BMIM-Cl through OmpF and to compare the calculated conductance to experimental data.

As a first step in the molecular modeling of BMIM-Cl, we compared theoretical and experimental values for the bulk conductivity in water. The simulations were performed using NAMD2.7[29] together with the CHARMM27 force field,[30] BMIM force field parameters,[31] and different water models, for example, TIP3P,[32] TIP4P,[33] TIP4P-Ew,[34] and TIP4P/2005.[35] Details are given in the Supporting Information. (Also see Figure 1b.)

The temperature-dependent bulk conductivity of 1 M BMIM-Cl was calculated for temperatures ranging from 10 to 60 °C using the approach of Aksimentiev and Schulten.[11] (Also see the Supporting Information.) As a control of the underlying interactions between

Table 1. Self-Diffusion Coefficient and Bulk Conductivity for Different Water Models Together with Experimental Values at 25 °C

data set	self diffusion $[10^{-5}\ cm^2/s]$	bulk conductivity [mS/cm]
experimental	2.27[36]	46.26
TIP3P	5.06[37]	97.07
TIP4P	3.29[37]	67.57
TIP4P-Ew	2.4[34]	49.51
TIP4P/2005	2.08[35]	38.46

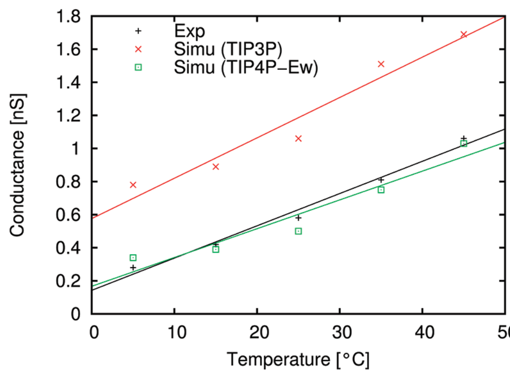

Figure 3. Pore conductance versus temperature for experiment and simulation using two different water models together with linear fits of the data sets.

ions and water, the temperature dependence of the calculated conductivities is compared with the experimental values. (See Figure 2.) As in previous simulations for KCl conductivity, we initially employed the TIP3P water model.[13,14] Although the slope of the simulation data with the TIP3P water model is very similar to the

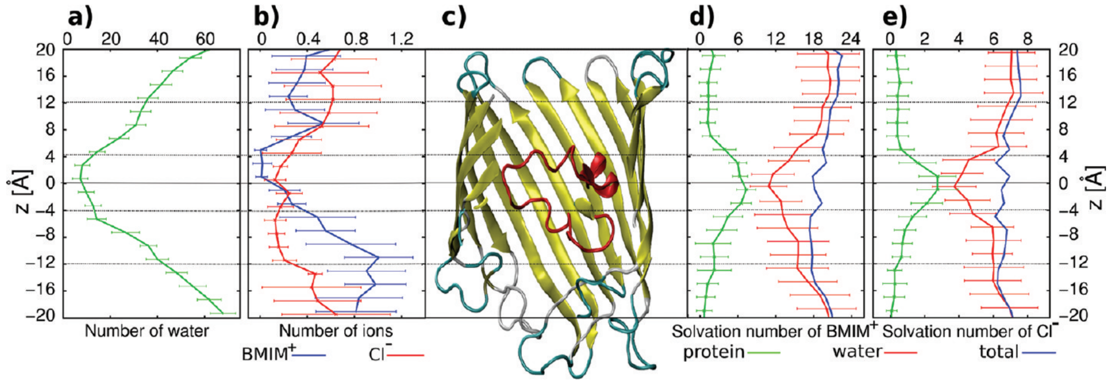

Figure 4. Several ion quantities as a function of the channel coordinate *z*. (a) number of water molecules; (b) number of ion molecules per species; (c) channel in the same scale as the ordinate; (d) solvation number of the $BMIM^+$ ions; and (e) solvation number of the Cl^- ions. (See the main text.) The dotted horizontal lines are given to help guiding the eye.

experimental results, a considerable parallel shift is visible. Surprisingly, the ratio between the calculated conductivity at room temperature and its experimental counterpart is almost equal to that of the corresponding self-diffusion constants of water as listed in Table 1. Motivated by the possible connection between conductivity and self diffusion, different water models were probed with self-diffusion constant in decreasing order. (See Table 1.). In conclusion, we found that the conductivity of 1 M BMIM-Cl solution is strongly correlated with the self-diffusion constant of the water model and that the best results are obtained for the TIP4P-Ew water model (Figure 2, Table 1). This is not too surprising because both quantities, the conductivity of BMIM-Cl and the self-diffusion constant of water, involve the process of moving a molecule, that is, either a $BMIM^+$ ion or a water molecule, through the surrounding water.

After the validation of the force fields and water models, the transport through OmpF was investigated. We performed 30 ns simulations for different temperatures and using the TIP3P and the TIP4P-Ew water models. (See the Supporting Information.) The current was determined as previously described.[11,13,14] The applied voltage of 1 V in the simulations is considerably larger than the experimental one to increase the number of passing ions during the simulation time, that is, to improve the statistics. The experimental results of the pore conductance of OmpF using 1 M BMIM-Cl solution at different temperatures show a linear dependence as shown in Figure 3. (See the Supporting Information for experimental measurement details.) Similar to the bulk conductivity calculations, the simulations using the TIP3P water model yield a considerably higher conductance, although also an almost linear increase with temperature. Results obtained with the TIP4P-Ew water model are in good quantitative agreement with the experimental results. These results further demonstrate the dependence of such conductance measurements on the water model and their self-diffusion constant. The good agreement between experimental and theoretical conductance can be further exploited to obtain atomic-level details of the transport process of BMIM-Cl through the OmpF pore.

In contrast with experiments, the simulations immediately yield the contributions of the individual ion types to the current, that is, $BMIM^+$ and Cl^-. (See also the Supporting Information movie.) The ratio of the Cl^- to $BMIM^+$ currents is found to be in the range of 3−5 increasing with temperature in the range from 5 to 45 °C (data not shown). Analysis showed that the mobility of Cl^- ions is influenced more by the increasing temperature than the $BMIM^+$ ions. This leads to an increased ratio of Cl^- to $BMIM^+$ currents at higher temperatures. Furthermore, it is found that, at 25 °C, $BMIM^+$ ions have an approximately four times higher residence time in the constriction region compared with the Cl^- ions (see also below).

For further analysis of the MD data at 25 °C, the channel was divided into bins of 2 Å width along the channel coordinate *z*. If the center of mass of the molecule under investigation was within the *z* range of this bin, the count of the individual quantities was increased. Subsequently, the number of molecules was averaged over the trajectory and are displayed in Figure 4. In fact, the number of water molecules can be used as an estimate for the channel size. As can be clearly seen, the constriction zone extends from roughly −4 to +4 Å. (See Figure 4a.) For the analysis of the average ion occupation in the channel, note that the applied positive voltage forces the Cl^- ions down toward negative values of the channel coordinate *z* (from the periplasmic side of the membrane to the extracellular side) while the $BMIM^+$ are dragged up toward positive values. Compared with the average water occupation in the channel, the Cl^- ions have a slightly higher average number above the constriction zone and a slightly depleted occupation at negative *z* values. (See Figure 4b.) This is due to the hindrance caused by the constriction region. Once the narrow part of the pore is passed, the ions are easily dragged away by the external field. As can also be seen in the Supporting Information movie, the effect is much more enhanced for the bulky $BMIM^+$ ions, which are crowded below the constriction zone.

For the present nonspecific channel, we examined the solvation characteristics of permeating $BMIM^+$ and Cl^- ions along the channel coordinate *z*. In general, ion channels regulate the flow of ions across the membrane via substituting the loss of hydration with favorable protein interactions.[38] To be able to quantify these, we determined solvation shell definitions of the $BMIM^+$ and Cl^- ions by calculating radial distribution function (RDF) for the previously described simulation of 1 M BMIM-Cl aqueous bulk solution and 4 ns long trajectories. For the $BMIM^+$ molecule, the center of charge was considered, and a first minimum was found at 4.1 Å, which was subsequently used as radius of the first hydration shell. Similarly for Cl^-, the radius of the first hydration shell was obtained as 3.2 Å. With regard to the protein contacts, only oxygen atoms were considered for the contribution of the protein to the solvation number of $BMIM^+$ ions and nitrogen atoms for that of Cl^- ions. For the ion−protein contacts, the same distance criteria as

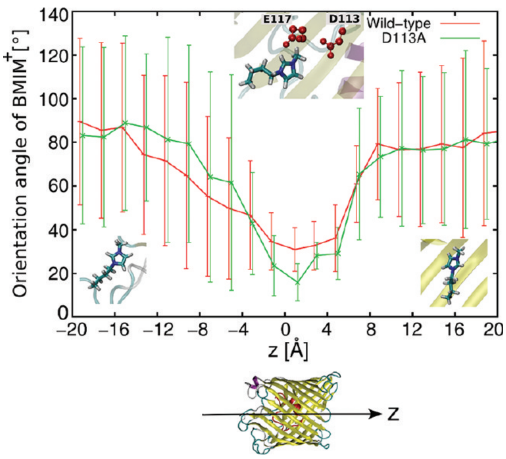

Figure 5. Average orientation of the $BMIM^+$ dipole with respect to the channel coordinate Z with error bars showing the standard deviation per bin. Representative orientations of $BMIM^+$ ions are shown in the insets for three particular regions of OmpF (center inset includes two important negatively charged residues in constriction region; see the main text).

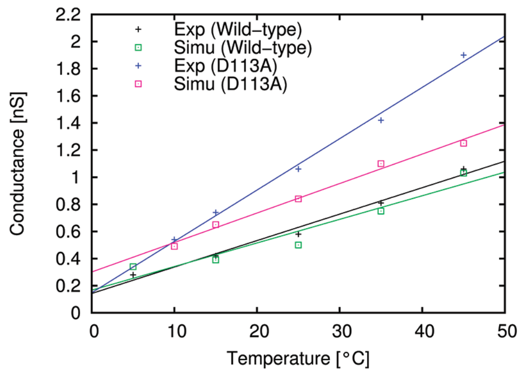

Figure 6. Experimental and calculated pore conductance for the D113A mutant. For comparison, the conductance data for the wild type are shown as well.

those for the ion–water contacts were applied. The hydration number of the ions progressively alters in response to the changing environment as they move through the OmpF pore (Figure 4d,e). The hydration number appears to be minimal in the constriction zone with approximately 10 to 11 water molecules around each $BMIM^+$ ion and 4 water molecules around each Cl^- ion. Interestingly, the loss of hydration is compensated by protein contacts in the constriction region. Apparently, the contributions of water and protein contacts complement each other to keep the total solvation number per ion roughly constant throughout the channel. For KCl ions in OmpF, a similar phenomenon was previously observed by Im and Roux.[27] In addition, both $BMIM^+$ and Cl^- follow separate pathways over the length of the pore in a screw-like fashion. (See the Supporting Information.)

Compared with the simple potassium and chloride ions, an important feature of the more bulky $BMIM^+$ ions is their orientation with respect to the channel. This is a key feature not accessible by experiment but by MD simulations. As described above for the other calculations, the channel was divided into bins of 2 Å width along the channel coordinate z and the quantities were averaged over the full length of a 30 ns trajectory. The orientation angle was measured between the dipole of each $BMIM^+$ ion residing in a particular bin and the z axis. As shown in Figure 5, the orientation of the $BMIM^+$ ions changes gradually as it passes through the pore. Toward the extracellular ($-20 \leq z \leq -12$ Å) and the periplasmic side ($12 \leq z \leq 20$ Å) of the pore, $BMIM^+$ has an average orientation perpendicular to the channel axis with very large values of the standard deviation. This perpendicular direction is due to the charged residues lining the pore walls. (See Figure S2 in the Supporting Information.) These charges create local electric field components perpendicular to the channel axis and the $BMIM^+$ ions partially align to them. At the same time, the large standard deviation of the orientation angle toward the extracellular and periplasmic side of the pore shows that the ions have a large flexibility in their movement. The situation is different near the constriction region. The pore diameter is not large enough for $BMIM^+$ ions to pass in a perpendicular orientation. Therefore, they progressively start to change their orientation as they approach the constriction region. There, the $BMIM^+$ ions align almost parallel to the channel axis. In terms of the angle between dipole moment and channel axis, this leads to values around 30°. As shown in the middle inset of Figure 5, the positively charged headgroup, that is, imidazole ring, of the $BMIM^+$ ion tends to tilt toward negatively charged residues in the constriction region, namely, D113 and E117. This leads to an average angle of the $BMIM^+$ dipole away from the expected 0°. Probably more important than the average angle is the relatively small standard deviation in the constriction zone. Although not directly visible in Figure 5 (but, e.g., in the Supporting Information movie), the $BMIM^+$ ions have to pass the constriction zone with a certain directionality, that is, the imidazole ring first. This shows that the ions first have to acquire a certain orientation before they can actually pass this region. On average, this leads to a prolongation of the passage time or a decrease in the current produced by the $BMIM^+$ ions. Figure 5 shows a second curve, the orientation of the $BMIM^+$ ions through the OmpF mutant D113A, where a negatively charged aspartate residue in the constriction zone was mutated into a neutral alanine. The same system setup as that for the wild type OmpF simulations was used with the TIP4P-Ew water model; then, the D113 residue was mutated. Subsequently, the system was equilibrated for 5 ns, followed by a applied-field simulations for 10 ns. Because of the removal of a negative charge in the constriction zone, one may expect a change in the average orientation of the $BMIM^+$ dipole in this region. This can be attributed to the reduced electrostatic force experienced by the positively charged imidazole ring leading to a less pronounced tilting of the $BMIM^+$ head. Indeed, the average angle in the mutant is found to be ~15° compared with roughly 30° in wild type. Farther away from the constriction zone, the average orientation and its standard deviation are rather similar to the wild type data. This is reassuring because it shows that these two independent simulations yield very similar results away from the constriction zone, as to be expected.

In Figure 6, the temperature-dependent conductance of the D113A mutant from electrophysiology and MD simulations is displayed. At the lower considered temperatures, the agreement between both approaches is good, but it is obvious that the experimental and the theoretical curves have different slopes. The curve from the simulation is parallel shifted from the wild type results to larger conductance values. In the case of the experiments, there is a considerable change in slope. Because of

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the very good agreement between theory and experiment for the wild type OmpF, these data are somewhat surprising. Although the results are within the range of expected agreement between experiments, it is obvious that some contribution is neglected in the mutant. Nevertheless, the structure of the mutant was quite stable in the simulations. Furthermore, previous studies of the conductance of simple ions provided good agreement with experiments.[13,14] The ratio of Cl^- to $BMIM^+$ currents was found to be in the range of 5−8 increasing with temperature. Compared with the ratio of 3−5 for the wild type, it was observed that mainly the Cl^- ions are responsible for the larger conductance of the D113A mutant because the contributions from the $BMIM^+$ ions were roughly the same as those for the wild type.

In conclusion, we have experimentally measured the temperature-dependent conductance of aqueous BMIM-Cl solution through OmpF trimer and also modeled it using MD simulations. An excellent agreement is obtained between electrophysiology and simulations of wild type OmpF and a reasonable agreement for the D113A mutant. In addition to the experimental findings, the simulations yield a molecular-level picture of the ion transport. In particular, the kinetic simulations reveal the rate-limiting interactions of the ions with the channel wall. As an important issue of the transport process, we have calculated the orientation of the $BMIM^+$ ions. It is really astonishing that one can see the realignment of the bulky ions within the time scale of the applied-field MD simulations. The necessity for the correct orientation of $BMIM^+$ near the constriction region may serve as a rate-limiting step of the passage, thereby slowing down the kinetics of antibiotic translocation through OmpF. This probably allows us to observe an increased residence time of ampicillin in OmpF with BMIM-Cl compared with KCl, as shown in ref 22. For the translocation of substrates such as antibiotic molecules, the translocation time is too long to be observed directly in standard MD simulations. Hence, one needs to go to more indirect schemes as the calculation of the potential of mean force or steered MD simulations to understand these processes theoretically. Nevertheless, insights from the present study can be further exploited to understand antibiotic translocation processes and to investigate other ILs for similar electrophysiological measurements through nanopores.

■ ASSOCIATED CONTENT

S Supporting Information. Experimental and computational details and movie of the ion transport simulations. This material is available free of charge via the Internet at http://pubs.acs.org.

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